段考必錯 #5 同類練習題：和角公式

高二｜三角函數不能像分配律那樣拆，要用和角、差角公式　姓名：_____

1. $\sin 75°$（提示：$75° = 45° + 30°$）＝_____

2. $\cos 15°$（提示：$15° = 45° - 30°$）＝_____

3. $\cos(45° + 30°) =$ _____

4. $\sin(60° - 45°) =$ _____

5. 已知 α、β 都是銳角，$\sin\alpha = \frac{3}{5}$，$\cos\beta = \frac{5}{13}$，求 $\sin(\alpha + \beta) =$ _____

解答

1. $\sin 75°$（提示：$75° = 45° + 30°$）$= \dfrac{\sqrt{6}+\sqrt{2}}{4}$

2. $\cos 15°$（提示：$15° = 45° - 30°$）$= \dfrac{\sqrt{6}+\sqrt{2}}{4}$

3. $\cos(45° + 30°) = \dfrac{\sqrt{6}-\sqrt{2}}{4}$

4. $\sin(60° - 45°) = \dfrac{\sqrt{6}-\sqrt{2}}{4}$

5. 已知 α、β 都是銳角，$\sin\alpha = \dfrac{3}{5}$，$\cos\beta = \dfrac{5}{13}$，求 $\sin(\alpha + \beta) = \dfrac{63}{65}$